

Classify abalone sex (M, F, I) based on morphologies

4176 entries,
9 morphs

Train: 3132 entries

Test: 1044 entries

male



female



infant



7 models to predict sex
based on morphs

Random forest works best on train set (error 45.02%), achieves error of 42.82% on test set

Train set

Model	Error Rate
Full CART model	0.505125
Pruned CART model	0.4693616
New model	0.4623461
Bagging	0.4626615
Random forest	0.4502096
Adaptive boost	0.4677622
XGBoost	0.6596508

Test set: err rate = 0.4282

Our random forest model was not overfitting or randomly guess

3 classes is not an easy task to do